GAS ABSORPTION

Pre-Lab Plan
Unit Operations Lab 5
3/17/2026

Group #1, Tuesday 1PM section

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1. Experimental Objective (5pts)

The experiment has two primary objectives: first, to determine fluid flow and pressure drop characteristics of a packed bed gas absorber, including the flooding point. Second, determine the gas absorption characteristics and overall mass transfer coefficient for the removal of CO_2 from an air stream using deionized water. The mass transfer expression, $N = ka\Delta C$, will be used to quantify the rate of CO_2 transport between air and water phases.

2. Experimental

In this experiment gas absorption will be investigated through a packed bed column for the removal of carbon dioxide (CO_2) from an air stream using water as the absorbing medium. The system operates under countercurrent flow conditions, with the gas phase flowing upward and the liquid phase flowing downward through a column packed with Raschig rings to promote interfacial contact and mass transfer of CO_2. Key experimental measurements include the pressure drop across the packed bed to identify flooding behavior, as well as the determination of CO_2 concentrations in both gas and liquid phases to evaluate absorption performance and overall mass transfer characteristics. Lab includes a liquid extraxtion with CO_2 and a titration with Sodium Oxide to determine the concentration of CO_2 at different counter current conditions.

2.1 Apparatus (5pts)

The system is an Armfield Gas Absorption Column (UOP7) packed 1-cm Raschig rings to facilitate contact between a falling water stream and rising air carbon dioxide gas mixture. Provided is a singular table detailing the various components of the apparatus. To coincide with it are four figures of pictures taken during the introduction of the lab day. Labeled on them are numbered labels associated with the components numbers of the table.

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Component No.	Component	Description
1	Gas absorption	Vertical transparent column packed with

	column	Raschig rings for gas-liquid contact.
2	Raschig ring packing	Provides high surface area for mass transfer inside column.
3	Liquid distribution section	Ensures even distribution of water over packing.
4	Liquid collection section	Collect liquid exiting the packed bed.
5	Liquid reservoir tank	Holds circulating water for the system.
6	Centrifugal pump	Circulates water from reservoir to top of column.
7	Electric motor	Provides power to the Centrifugal Pump.
9	Gas rotameters (C1 and C2)	Measure flowrates of air and CO_2.
10	Liquid rotameter (C3)	Measures water flowrate.
11	Control valves (C1 C2 C3 C4 C5)	Regulate gas and liquid flow rates throughout the apparatus.
12	Pressure regulators (on gas cylinder)	Control pressure of CO_2 supply.
13	CO_2 gas cylinder	Supplies carbon dioxide to the system.
14	Air supply line	Provides compressed air to the system.
15	Pressure gauges	Measure gas pressure at regulator and system.
16	Manometers	Measures pressure drop across the packed bed.
17	Sampling port	Allow collection of gas and liquid samples.

18	Drain valve (F1)	Allows for the drainage of liquid.
19	Hemple gas analysis apparatus	Measures CO_2 concentration in gas stream.

Figure 1

Figure 2

Figure 3

Figure 4

2.2 Materials and Supplies (5pts)

Material/Supplies	Description	Material or Supply
Item	**Purpose**	**Category**
Carbon dioxide (CO_2) gas	Solute gas for absorption experiment	Material
Compressed air	Air to conduct the experiment.	Material
Deionized water	Liquid absorbent phase	Material
Sodium hydroxide (NaOH) solution	Titrant for CO_2 analysis	Material
Phenolphthalein indicator	Indicates titration endpoint	Material
Raschig rings (1 cm)	Packing material for mass transfer	Material
Gas sampling syringe	Collects gas samples	Supply
Tubing and connectors	Transfers fluids between components	Supply
Graduated cylinder	Measures liquid volume / void fraction	Supply
Erlenmeyer flask (125 mL)	Holds samples for titration	Supply
Bucket	Collects drained liquid	Supply
Funnel	Assists in transferring liquids	Supply

2.3 Experimental Plan (10pts)

<u>PART 1</u> – Characteristics of Liquid and Gas Flow in Packed Columns

1. **<u>SAFETY</u>**: review MSDS on NaOH, phenolphthalein, compressed air, and CO_2. Gloves and goggles are necessary. Check the safety requirements for operating

high pressure gas cylinders. Treat all glassware with care and beware of slippery beakers/flasks.

2. Begin by calculating the void volume of the packed bed material.
    a. Do this by first adding a volume of bed material into a graduated cylinder. Then pour water until the voids of the bed material are filled.
    b. Divide the volume of the bed material by the volume of water added.
3. Next, fill the tank reservoir three-quarters full of DI water. Be sure to record the temperature of the water and to also close the drain valve located underneath the reservoir. Be sure that the column drain valve (D1) is also closed.
4. Next, prepare the apparatus for airflow by opening the compressed air shut-off valve (C2) and adjusting the air flowrate on flowmeter F2 to 20 L/min by using the control valve (C5). Ensure valve C3 is closed and then allow air to flow through the system for five minutes to dry the packings.
    a. Once packings are dry, measure and record the pressure drop (using the manometer) across the packed bed for different gas flowrates.
    b. No water is used in this step. Once data has been recorded, airflow can be temporarily shut off by closing valve C5.
5. Now add water to the system by flipping the red switch that starts the centrifugal pump. Open valve C4 and then C1 to expel air from the pump. Adjust the water flow on flowmeter F1 to 1.5 L/min by manipulating valve C1. Adjust valve C4 so that water level below the column is above the sample valve S4 but below the air inlet tube.
    a. **<u>NOTE:</u>** opening valve S4 and placing a bucket under the sample tubing to collect water will help maintain the proper water level.
6. Open valve C5 and adjust the air flowrate on flowmeter F2 to 20 L/min. Measure the gas pressure drop as you steadily increase the air flowrate until flooding within the column occurs. Repeat this step for three more water flowrates (3 L/min, 6 L/min, and 8 L/min).
    a. **<u>NOTE</u>**: when the pressure drop becomes large, or the pressure difference steadily increases at a given air flowrate, this indicates that you are at or quickly approaching the flooding point. Lower the air flowrate at this point. This will prevent any red oil from entering the tubing.

<u>PART 2a</u> – Absorption of CO_2 by H_2O in a Closed-Loop Water System

1. Collect a water sample and determine its initial CO_2 concentration by using titration paper.
    a. Do this by using a graduated cylinder and collecting 50 mL out of valve S4 (or reservoir tank if needed). Purge the line before collecting.

b. Pour the sample into a 125 mL Erlenmeyer flask and add 3-5 drops of phenolphthalein indicator solution.
c. If the solution turns red immediately this means, there's no free CO_2 present.
d. If the solution remains colorless, swirl the flask and titrate the solution with standard (0.01 N or 0.05) NaOH solution. The end point is reached when a 'definite' pink color persists for ~30 seconds.
e. Record the total volume of NaOH added.
f. Use equation: $C_d\left(\frac{gram\,mole}{liter\ of\ free\ CO_2}\right)=\left(\frac{V_B\times(Conc.of\ Base)}{mL\ of\ Sample}\right)$ to determine the amount of free CO_2 in the sample.
    i. $C_d = concentration\ at\ inlet, S6(the\ top\ of\ the\ column)$.
    ii. $C_{do} = concentration\ at\ the\ outlet, S4(bottom)$.
g. **NOTE**: solubility of CO_2 is a function of temperature and pH.
h. **NOTE**: all titration waste from this experiment should be placed in the titration waste bottles.

2. Once initial CO_2 concentration is found, place a cover over the tank.
3. Ensure valves C5 and C3 are closed. Then, open valve C4 and start the centrifugal pump. Open valve C1 a couple turns for 15-30 seconds to expel air from the pump, then adjust the water flowrate to 6 L/min on the flowmeter F1 by adjusting valve C1. Adjust valve C4 so that the water level below the column is above valve S4 but below the air inlet tube.
    a. **NOTE**: opening S4 and placing a bucket under the sample tubing to collect water will make maintaining the water level easier.
4. Next, allow 20 L/min of air flow through the system by opening the shut-off valve in the back of the unit and then manipulating valve C5. Make sure that valve C3 is closed. Be sure to record the pressure.
    a. Keep an eye on the air flowrate on flowmeter F2. Readjustments may be needed as the output tends to drift. Note down any adjustments that had to be made.
5. Now, carefully open the main valve on the CO_2 cylinder all the way. Slowly adjust the pressure regulating valve until a gauge reading of 20 psi is acquired, then open the valve just after the pressure regulator and heater block. Adjust valve C3 to give CO_2 a flowrate of 2 L/min through 5 L/min. Turn on the heater block switch (next to the pump switch).
6. After five minutes of operation, take 50 mL water samples at periodic intervals from valve S4 until the dissolved CO_2 values are constant. Analyze the samples through titration. Follow the same procedure stated in step 1.

7. Once the CO_2 levels have proven constant in the water (or 15 minutes have passed), using the Hempl apparatus, measure the volume fraction of CO_2 in the air leaving the column from sample port S2. Detailed steps are listed below.
    a. Start by filling the two globes on the left-side of the apparatus with 1.0 N NaOH solution.
    b. Pull the piston of the syringe all the way to the stop. Grab a squirt bottle filled with DI water and wet the piston. This ensures there's a water seal so the gas cannot leak out. This may have to be done several times throughout the entire experiment.
    c. Next, flush the sample lines by repeatedly sucking gas from the line, using the syringe, and expelling the gas into the atmosphere. (volume of the syringe is ~100 mL)
    d. Measure the diameter of the length of tubing that leads to the syringe. This will allow the calculation of the total volume of tubing. Flush sample line until the total volume has been emptied.
    e. After flushing and 15 minutes of steady operation, take gas samples from sample point S2.
    f. To do this, first isolate the absorption globe and close the vent to the atmosphere. Next, select a line and fill the syringe by pulling the piston outward slowly. Draw ~40 mL of gas. Let two minutes pass to allow the gas temperature to equilibrate with the temperature of the syringe.
        i. **<u>WARNING</u>**: if the CO_2 concentration in the sample gas is greater than 8%, it's possible to suck liquid into the cylinder. This will ruin the experiment and takes a considerable amount of time to fix. If this occurs, do not fully retract the piston. Stop it at a particular mark on the 'coarse' scale and the read from the 'fine' scale.
    g. Isolate the syringe from the column and absorption globe. Then vent the syringe to atmospheric pressure by opening the vent to the atmosphere. After ten seconds, close it.
    h. Connect the syringe back to the absorption globe. The liquid level shouldn't have changed. If it did, briefly open the vent to the atmosphere.
    i. Wait until the level in the indicator tube is zero. This means that the pressure in the cylinder is atmospheric.
    j. Slowly close the piston and empty the gas in the syringe into the absorption globe. Now, slowly draw the piston out again to prevent any gas leakage. Take note of the level in the indicator.
    k. Repeat step j until there is no significant change in the level indicator.
    l. The indicator tube marking, V2, represents the volume of the gas sampled.

8. If there's time, follow the 'steps to become familiar with the Hempl apparatus' mentioned in Appendix A to become confident in measuring the concentration of CO_2 in a gas mixture.

<u>PART 2b</u> – Desorption of CO_2 from H_2O into air

1. Once part 2a is complete; shut off the CO_2 flow using valve C3, decrease the liquid flow rate to 1.5 L/min using valve C1, open drain valve D1 under the column, and close control valve C4.
2. Adjust valve D1 to maintain a liquid level above S4, but below the air inlet tube.
3. Now, very quickly begin to take water samples (~1-minute intervals) from S4. Analyze the samples using the same procedure mentioned in part 2a (step 1).

<u>Part 3</u> – Absorption of CO_2 by H_2O in an open-loop water system

1. Close drain valve D1. Fill the reservoir back to three-quarters full of DI water. Record the temperature of the water. Also, collect a water sample to determine the initial CO_2 concentration following the procedure in part 2a, step 1.
2. Open control valve C4. Ensure valves C3 & C5 are closed. Then start the centrifugal pump; open valve C1 a couple turns to expel air from the pump (15-30 s), then adjust the water flowrate to 1.5 L/min.
3. Open the drain valve D1 and close valve C4. Using D1, maintain a liquid water height below the air/ CO_2 inlet but above the sampling valve S4. Place a bucket under S4 so it can continually drain to help maintain the desired level.
4. Open valve C5 and adjust the air flowrate on flowmeter F2 to 20 L/min. Monitor the level in the flowmeter as the reading tends to drift.
5. Next, carefully open the main valve on the CO_2 cylinder all the way. Slowly adjust the pressure regulating valve on the CO_2 regulator to attain 20 psi. Then, open the valve located just after the pressure regulator and heater block. Manipulate valve C3 to give the desired CO_2 flowrate. This lab will use flowrates 2-6 L/min. (five total CO_2 flowrates). Lastly, turn on the heater block (next to pump switch).
6. As soon as CO_2 is turned on, begin to take 50 mL water samples from S4 at 30-second intervals during the first two minutes. This interval may be increased to five minutes until the concentration reaches a steady state. Analyze the samples following the procedure in part 2a, step 1.
7. Analyze one gas sample from S2 to measure the volume fraction of CO_2 in the outlet air once steady state is reached using the Hempl apparatus.
8. **<u>NOTE</u>**: steps 2-4 need to be done as quick as possible. Or the flowrates pre-set, the reservoir again filled with DI water, and the experiment started to ensure there's enough water to complete the experiment.

Independent/Controlled Variables: gas flowrate (air + CO_2), water flowrate (via rotameter), pump operating power, gas composition at inlet, gas-to-liquid ratio, packed-bed material, column height/diameter, temperature within the column, and NaOH solution concentration.

Dependent Variables: flooding point, pressure drop across the column, flow characteristics within the apparatus, gas and water outlet CO_2 concentration, mass transfer rate, absorption efficiency, and the pressure withing the column.

POST-LAB PLAN:

1. Create plots of pressure drop vs. water and air flowrate to show packed bed characterization and to describe the various regimes of gas-liquid absorbers.
2. Demonstrate how the porosity of the bed material was calculated.
3. Make plots of CO_2 concentrations vs. time. Discuss the CO_2 concentrations initially in the air and at the outlet of the column.
4. Make a table of the calculated mass-transfer coefficients. This data will be used to describe the absorption and desorption process in the apparatus.

Project Safety Evaluation (10pts)

Project Title: GAS ABSORPTION

Date: 3/17/2026

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: Water dripping from the column during operation or water spilling when taking samples could possibly land onto electrical components below and cause short-circuiting. Caution must be applied and care must be used to clean all wet spots and not have any wires or sockets exposed.
Potential mechanical hazards?	Yes	If yes, identify conditions: Running the pump dry can damage it.
Potential pressure hazards?	Yes	If yes, identify conditions: Using a high-pressure CO_2 cylinder can cause damage to equipment if the regulating valve isn't operated slowly and freezing occurs due to the Joule-Thomson effect.
Potential temperature hazards?	Yes	If yes, identify conditions: Any direct contact with the heater block or gas streams that have had their temperatures adjusted.
Hazardous/toxic chemicals involved?	Yes	If yes, hazards involved: When using NaOH as an absorbent in the Hempl extraction of a liquid sample and when used as a titrant when analyzing the concentration of CO2 in the sample collected.
Gloves required?	Yes	If yes, specify type: Nitrile gloves must be worn when handling the NaOH solution to prevent chemical burns.
Other hazards	Yes	If yes, specify conditions: Potential spills of NaOH on glassware surface make the glassware slippery when coated with NaOH.
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: • Sodium hydroxide (NaOH) — **CAS: 1310-73-2** • Carbon dioxide (CO_2) — **CAS: 124-38-9** • Phenolphthalein — **CAS: 77-09-8**
Process Parameters	*Max Expected*	*List location and possible hazards and precautions*
Temperature (ºC)	100℃*1	**Location:** Heater block/Liquid reservoir. **Hazards:** Burn hazard from the heat coming from the heat block.

		Precautions: Carefully operate around the area. Especially when around the reservoir.
Pressure (psia)	1850 PSI *2 20 PSI	**Location:** Pressurized CO_2 cylinder **Hazards:** Pressure hazard that can cause valve damage and other mechanical failure related to the vessel and/or the apparatus. **Precautions:** Fully open the cylinder valve but slowly adjust the regulator.
Safety Controls	*yes/no*	*If yes, Provide description*
Pressure Gauges	*Yes*	*Pressure Gauges on the Air supply and* CO_2 *Gas Cylinder monitor and regulate the pressure of gas flowing within the apparatus.*
Waste Bottles	*Yes*	*These are containers in the fume hood used for the disposal of titration waste.*
Heater Block	*Yes*	*This regulates the temperature of the* CO_2 *stream and stops the regulator icing,*
Manometer	*Yes*	*The manometer gives us the pressure drop and indicates when flooding is about to occur in the column.*

*1 100℃ is an assumed value due to no information provided in the lab manual when consulting the max temperature found in the lab. Due to no information on the max temperature expected is grabbed from previous labs as a reference and Heavy heat Safety protocol will still be used during the operation of the lab.
*2 1850 PSI is the max pressure within the CO_2 Gas Cylinder.
*3 20 PSI is max pressure flowing through the apparatus.

SAFETY STATEMENT:

I have read relevant background material for the Unit Operations Laboratory entitled: "GAS ABSORPTION" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance with standards and acceptable safety practices and will conduct all my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Team Member Signature: Cesearo Dilosa — Date: 17 March 2026

Team Member Signature: Ify Bidokwu — Date: 17 March 2026

Team Member Signature: Joshua Laforest — Date: 17 March 2026

Team Member Signature: Javier Silva — Date: 17 March 2026